

10	(a) sharpness: how well the edges (of structures) are defined contrast: difference in (degree of) blackening between structures	B1 B1	[2]
	(b) e.g. scattering of photos in tissue/no use of a collimator/no use of lead grid large penumbra on shadow/large area anode/wide beam large pixel size <i>(any two sensible suggestions, 1 each)</i>	B2	[2]
	(c) (i) $I = I_0 e^{-\mu x}$ ratio = $\exp(-2.85 \times 3.5) / \exp(-0.95 \times 8.0)$ = $(4.65 \times 10^{-5}) / (5.00 \times 10^{-4})$ = 0.093	C1 C1 A1	[3]
	(ii) <i>either</i> large difference (in intensities) <i>or</i> ratio much less than 1.0 so good contrast	M1 A1	[2]
	<i>(answer given in (c)(ii) must be consistent with ratio given in (c)(i))</i>		

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